



Analysis and prediction of water quality using deep learning and auto deep learning techniques

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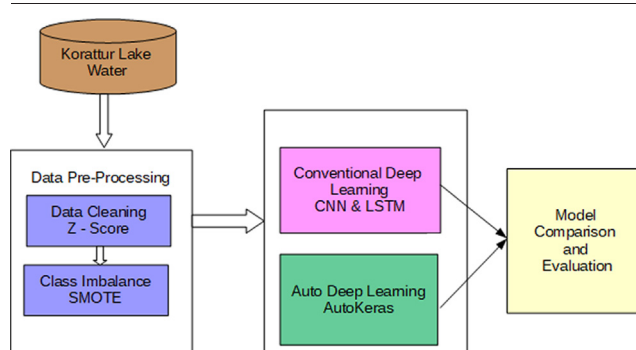
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HIGHLIGHTS

- Auto deep learning concept was adopted to analyze the water quality.
- Conventional and auto deep learning models were exploited for water quality analysis.
- The accuracy of conventional DL is 1.8% higher than that of AutoDL for binary class water data.
- The accuracy of conventional DL is 1% higher than that of AutoDL for multiclass water data.
- Conventional DL models outperform AutoDL model for this water data.

GRAPHICAL ABSTRACT



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ABSTRACT

Natural water sources like ponds, lakes and rivers are facing a great threat because of activities like discharge of untreated industrial effluents, sewage water, wastes, etc. It is mandatory to examine the water quality to ensure that only safe water is available for consumption. Traditional methods of water quality inspection are a cumbersome process and hence, Artificial Intelligence (AI) can be used as a catalyst for this process. AutoDL is an upcoming field to automate deep learning pipelines and enables model creation and interpretation with minimal code. However, it is still in the nascent stage. This work explores the suitability of adopting AutoDL for Water Quality Assessment by drawing a comparison between AutoDL and a conventional models and analysis to foresee the quality of the water, an appropriate class based on Water Quality Index segregating water bodies into different classes. The accuracy of conventional DL is 1.8% higher than that of AutoDL for binary class water data. The accuracy of conventional DL is 1% higher than that of AutoDL for multiclass water data. The accuracy of conventional model was ~98% to ~99% whereas AutoDL method yielded ~96% to ~98%. However, the AutoDL model ease the task of finding the appropriate DL model and proved better efficiency without manual intervention.

1. Introduction

Poor water quality affects all living organisms consuming it. A polluted water body can cause a series of imbalances that can have fatal

consequences in many instances. The water quality can be assessed to determine its purity. Conventional practices are labour-intensive and time-consuming, which demands to develop an automated model. Water data considered in this research work includes quantities like Total Dissolved

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Salts, pH, Phosphate, Turbidity, Iron, Nitrate, Chloride, Sodium and Chemical Oxygen Demand (COD) from Korattur Lake in the Chennai city. Based on these parameters, the motivation of this work is to build an AutoDL model and compare it with a model built using conventional deep learning pipeline and thereby, analyze the suitability to adopt AutoDL to predict water quality effectively.

1.1. AutoDL

Automated DL (Deep Learning), being a subset of Auto ML, is used as a library underneath it. As the name suggests, it automates the Deep Learning pipeline from data selection to constriction of layers to output classification. Reducing human intervention, it brings deep learning out of the bounds of data scientists and provides universal access for usage.

By the developers of DATA Lab located at the Texas A and M University as well as other contributors of the community, an open-source software library was formed for AutoML (ML - Machine Learning), namely, AutoKeras. The so-called “Keras models” have been the means of conducting the search utilizing TensorFlow tf.keras API. AutoDL aims at the repetitive data processing and learning tasks and helps automate the procedures without human intervention, and if one may venture to say, optimized efficiency. It is one step towards intelligent machines that can recognize, understand and perceive the nature of input and process appropriately.

1.2. AutoDL in context

AutoDL model finds appropriate highly efficient models automatically spread across various predictive modelling tasks. It is a good candidate for the nature of data being dealt with in Water Quality Analysis, and helps draw parallels against the traditionally built Deep Learning models as in our case. A socially sensitive project such as water quality management must employ the algorithm of highest efficiency with scope for automation considering the range of application.

1.3. Motivation and purpose

The proposed work derives its motivation from two aspects. Social betterment and technological furtherment. Drawing upon the first aspect, water quality analysis is undeniably a major player in the public health sector. Increasing population and economic crisis are sister problems that hinder water quality management. This problem might snowball into rampant spread of diseases and a general drop in the standards. Hence making the issue crucial and there is a need for further consolidation and advancement.

Secondly, while research continues to surge in this sector, it is fractured in individual tracks. While individual algorithms and parameters are analyzed to sufficient depth, there has been no comparative study between the same. A contextual approach with a comparisational narrative that cumulates all the explored algorithms and explores scopes for development is crucial. The importance of this can be viewed in the light of the limited data available. DL techniques require proper data and hence probing into data engineering techniques becomes inevitable. This may improve the results of the traditional Deep Learning techniques available in hand.

Taking a step further, this paper explores Automated Deep Learning, which is a new research domain. This paper straddles perfectly built DL models and automated DL models. With the same baseline data, the authors intend to explore the potential of automated processing, as well as the shortcomings in it. Moreover, the paper attempts to understand how an automated model interprets the data and the nature of the model it considers efficient. Hence the purpose lies in comparison of the performance of traditional and automated models and their parameters and examination of social relevance of Automated Water Quality Analysis.

2. Literature survey

The main scope to be displayed by this research is foretelling the water quality to make sure that water is drinkable. Using traditional methods,

analysis of water quality is conducted in laboratories that are not accurate and consumes lot of time. Therefore, this work carries out DL methodologies in eradicating this difficulty. In the work proposed by in the work (Prasad et al., 2020, 2021a), Korattur Lake, Tamil Nadu was the source for water sampling, and these samples were checked by making use of various ML as well as DL models like RNN, LSTM as well as ANN. LSTM was found to be best suitable algorithm among the three DL algorithms having the highest precision of 94%.

Performance of the model was drastically improved while predicting the water quality with multiple neural networks compared with a single neural network as proposed by Ahmad et al. (2017). The work of Solanki et al. (2015), utilized time series data taken via USGS National Water Information System (NWIS), highest accuracy in foretelling the WQI and KNN, Naive Bayes as well as SVM were used to predict water Quality Classification seemed to be achieved when the data set was trained using ANN-NAR. According to the research in Randrianiaina Jerry et al. (2019), deep learning was used to process the quality parameters such as pH, DO, turbidity as well as conductivity modelling at Itasy Lake. Performance exhibited by hybrid CNN-LSTM model together seemed to be more efficient than CNN and LSTM as individual models, taken from Small Prespa Lake in Greece (Barzegar et al., 2020; Chen et al., 2020; Yan et al., 2021; Li et al., 2019). The drawback of these methods was not enough characteristic quantities of the quality of water, not enough accuracy, incapable in handling many dimensions as well as unbalanced datasets (Kang et al., 2017). Work mentioned in Yang and Shami (2020) and Ye et al. (2019) identifies proper hyper parameters configurations effectively. The NARNET (nonlinear autoregressive neural network) as well as LSTM were utilized (WQC), resulting in NARNET and SVM championing their respective tasks (Aldhyani et al., 2020; Gocic et al., 2016; Haghiabi et al., 2017). The water quality prediction using LSTM DNN in IoT environment was discussed in the works (Liu et al., 2019; Wang et al., 2017; Bi et al., 2021). The work carried out in Su et al., (2020) was a wonderful initiative that acknowledged and addressed the lack of automation in water quality prediction reinforcing the gravity of the objective (Searcy et al., 2018). Water Quality Index (WQI) is computed using Weighted Arithmetic Index. Principal Component Regression was utilized to select important water parameters and Gradient Boosted Classifier was applied to predict water quality (Khan et al., 2021). With respect to AutoML, Alexandru-Ionut Imbrea's (Imbrea, 2021) work attempts to sum up some of the issues regarding adoption of off-the-shelf AutoML tools for data streams and proposed a meta learning approach that has the potential to be the core of an AutoML tool for data streams (Prasad et al., 2021b). The proposed benchmarking framework can be used for building AutoML tools that are more effective and efficient.

The work discussed in Escalante (2020) deals with the historic progression of Auto ML with survey on various ML techniques. The review on Automated Machine Learning techniques with respect to pipeline, including feature engineering, data preparation, neural architecture search (NAS) as well as hyperparameter optimization (He et al., 2019; Wen et al., 2020). The work carried out in Zöller and Huber (2021) applied chosen Automated Machine Learning frameworks taken for 137 data sets via well initiated AutoML identified suits. Review given explain the mathematical formulations, covering pipelines as well as other open-source options, is present in Li et al. (2018) works. The work in Elshawi et al. (2019) throws light upon the potential areas that could be improvised with respect to AutoML adjusting CASH (Combined Algorithm Selection and Hyper Parameter Tuning) problem. The work (Yao et al., 2018) reviews the existing AutoML works which gives us perceptions into fruitful applications (Choi et al., 2021). The Machine learning and Deep Learning algorithms were also used for prediction of wastewater treatment quality. Long Short-Term Memory model could predict the concentration of wastewater parameters. Pre-processing techniques could improve the accuracy of the model by selecting only the important chemical nutrients for further prediction process. One-hot encoding and Sliding Window techniques were used to pre-process the data and LSTM was used to predict the anomalies in the wastewater treatment prediction (Farhi et al., 2021). The accuracy obtained using these techniques was ~90% to ~99%.

3. Methodology

The methodology used in this research work is depicted in the following Fig. 1. Korattur lake water data was obtained and pre-processed using Z-score normalization and SMOTE to deal with class imbalance in the original dataset. Conventional deep learning methods such as CNN & LSTM are explored to predict the water quality. The proposed work in this research involved AutoDL technique which was implemented using AutoKeras library. The constructed models were evaluated and analyzed using metrics such as accuracy, precision and recall. The results of conventional DL and AutoDL models were compared, and results were tabulated.

3.1. Dataset collection

The dataset has been collected from Chennai Metrological department of Korattur Lake. The lake spans over 990 acres, supplying public drinking water for over eighteen years to the Chennai city. The dataset provides water samples from the year 2009 to 2019 which includes 5000 records with 9 parameters. The parameters are Turbidity, TDS, COD, PH, Phosphate, Iron, Nitrate, Sodium and Chloride.

3.2. Dataset description

The distribution of both two-class and more than two classes can be seen in Figs. 2 and 3. Table 1 depicts the Korattur lake water data description used for binary and multi class classification.

3.3. Data splitting

Prior to coaching the DL model, there is a requirement to classify the data as testing as well as training sets. Training and test sets were taken in ratio, 4:1. That is, for a sample set of 5000 numbers, 4000 were utilized as the training set and the rest 1000 set were utilized as testing set.

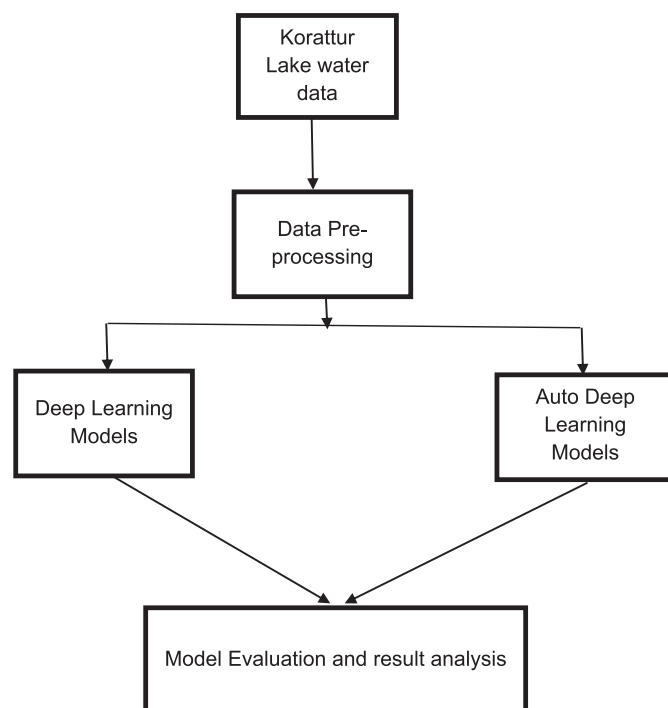


Fig. 1. Architecture diagram.

Binary-class Dataset

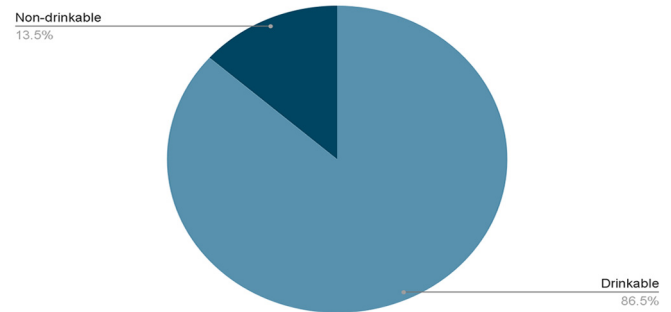


Fig. 2. Distribution of binary-class dataset.

3.4. Water Quality Index (WQI)

Using the listed nine characteristic quantities like the Total dissolved salts, pH, Phosphate, Turbidity, Iron, Nitrate, Chloride, Sodium and COD, the WQI is computed. Formula utilized in computing the WQI is discussed in the work (Prasad et al., 2020). These are the influential parameters which affects the water quality. Refer the paper for chemical parameters and Indian Standards (WHO, n.d.; Central Pollution Control Board, n.d.). Weights are assigned to each chemical parameter based on their degree of impact on the water quality. The weights are assigned from 1 to 5, 5 being the high degree of impact on water as given in Table 2. Based on WQI, water is categorised into various classes as seen below in Fig. 4 in which the Y axis represents the value of WQI.

3.5. Data preprocessing

3.5.1. Data cleaning

There are no missing values and no or very few outliers in the input data and hence the data is noise-free. This does not demand any outlier removing techniques.

3.5.2. Scaling data

The Standard scaling technique is utilized in scaling features. This method scales all the values that centre surrounding the mean and having unit standard deviation. The formula below is used for standard scaling:

$$z = \frac{(x - \mu)}{\sigma} \quad (3)$$

where μ = mean and σ = standard deviation, Z is the scaled (normalized) value and x represents the value of a particular sample.

Multi-class Dataset

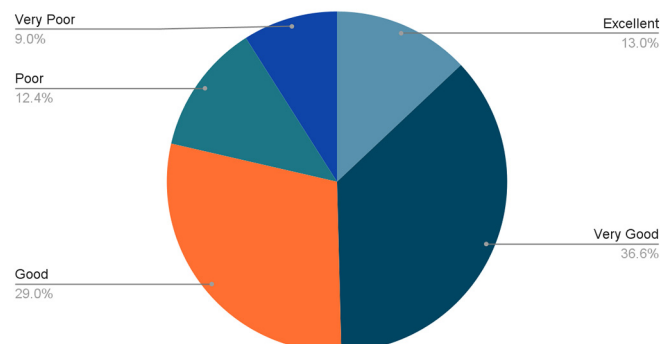


Fig. 3. Distribution of multi-class dataset.

Table 1

Description of the dataset.

Name of the lake	Records available quantity	Total parameters	Classes nos.	Class distribution
Korattur Lake	5000	9	2	Non-drinkable - 675 Drinkable - 4325
	5000	9	5	Very poor - 450 Poor - 620 Good - 1450 Very good - 1831 Excellent - 649

Table 2

Impact of chemical parameters on water quality.

Sl. No.	Chemical parameter of water	Weights assigned
1	Sodium & chloride	5
2	EC, TDS	4
3	Sulphate and COD	3
4	Calcium, magnesium	2
5	Iron, nitrate, phosphate and pH	1

3.5.3. Balancing classes

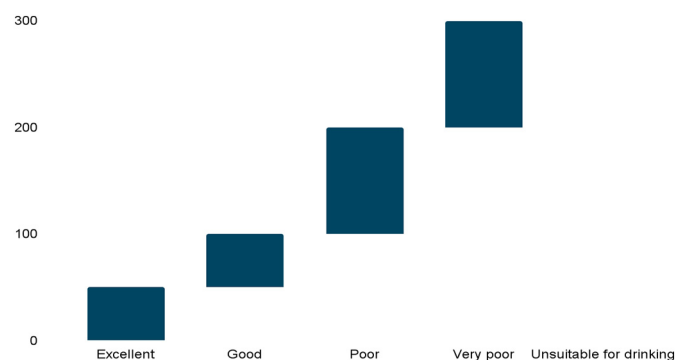
The dataset is highly unbalanced and the data set is balanced utilizing an oversampling method, called Synthetic Minority Oversampling Technique (SMOTE). The SMOTE technique for balancing the dataset is discussed in the work (Prasad et al., 2020; Chawla et al., 2002).

4. Methods

The below mentioned deep learning techniques are used in computing the water sample data taken from the lake and for classifying the models. The CNN, LSTM and AutoKeras models were exploited for the study of water quality prediction. CNN and LSTM are manual deep learning techniques whereas AutoKeras is an AutoDL library. There is no data transition between manual and auto deep learning model. The same data is used for study both conventional and auto DL models.

4.1. CNN

CNNNet (LeCun et al., 2015) is a multi-layered, feed-forward neural network. It uses one or more convolutions in its layers. The layers can be input, output and hidden layers. While the function of input and output layers is self-explanatory, a hidden layer usually performs multiplication or dot product. One can build various layers like fully connected layers, normalization layers and pooling layers (Baek et al., 2020). There exists an activation function that triggers a neuron to output a value, i.e., if it exceeds a certain threshold. It is usually ReLU activation function which outputs zero if input is negative and provides the input itself when non-negative. An example of a CNN model structure can be found in Fig. 5.

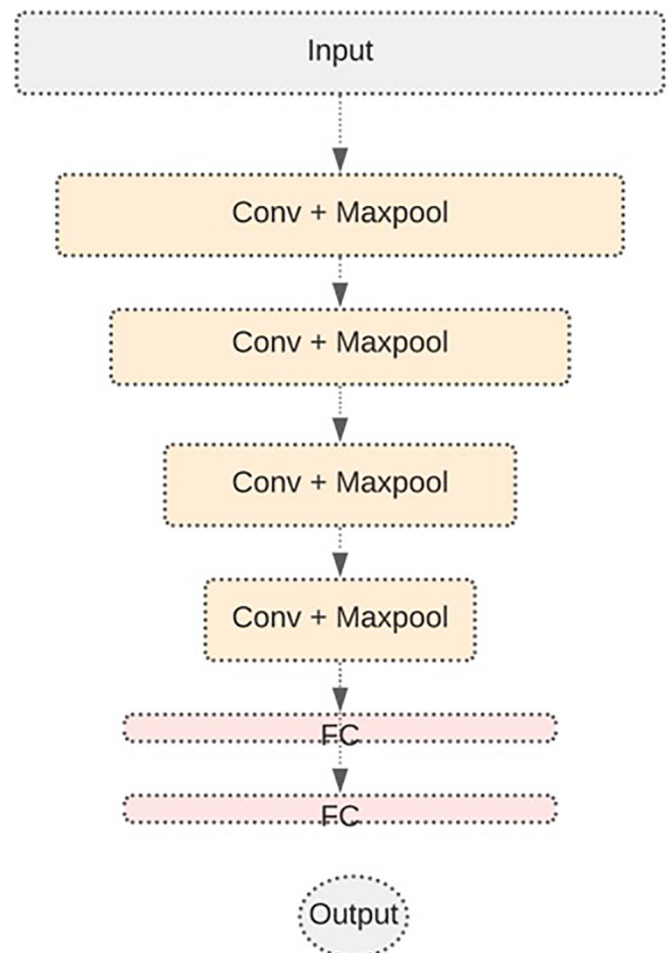
**Fig. 4.** Quality of water on the basis of WQI.

4.2. LSTM

In LSTM (Long Short Term Memory) (Hochreiter and Schmidhuber, 1997) the information flows through different cell states. This enables LSTMs to remember or forget things selectively (Hu et al., 2019). The information in a cell state has the following dependencies: the cell state formerly, current input as well as the former hidden state. The machine operates the following gates: the forget gate that erases data from the previous stage, the input gate that stores newly obtained information and the output gate that predicts the output based on the given input (Farhi et al., 2021; Khullar and Singh, 2021). Fig. 6 shows the LSTM architecture.

4.3. AutoKeras

By the developers of DATA Lab located at the Texas A and M University as well as other contributors of the community, an open-source software

**Fig. 5.** Convolutional neural network.

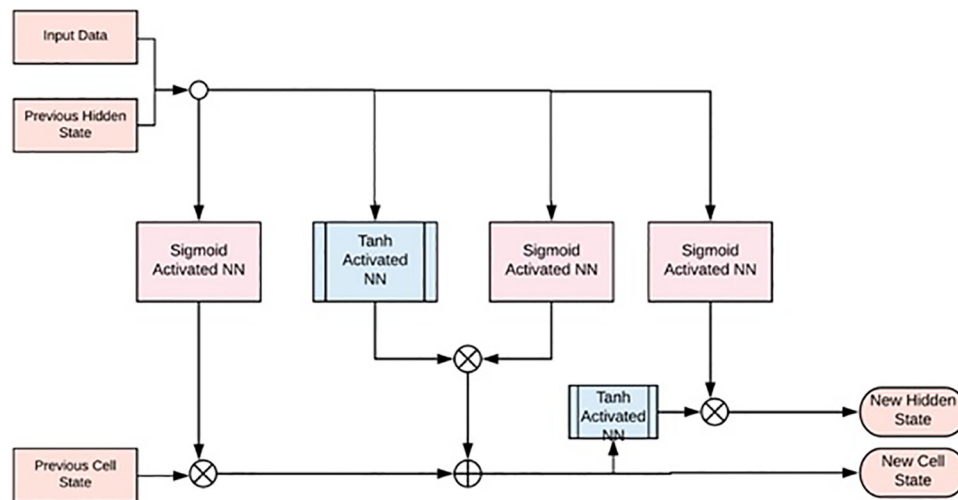


Fig. 6. Long short term memory.

library was formed for AutoML (ML - Machine Learning), namely, AutoKeras. The so-called “Keras models” have been the means of conducting the search utilizing TensorFlow tf.keras API. It finds appropriate highly efficient models automatically spread across various predictive modelling tasks, covering regression datasets as well as tabular (the structured classification). It is a good candidate for the nature of data being dealt with in Water Quality Analysis and helps draw parallels to the traditionally built Deep Learning models. The metrics used to evaluate the deep learning and Auto DL are Accuracy, Precision and Recall (Prasad et al., 2020). The actual formulae to calculate the metrics are given in the work (Ahmed et al., 2019).

5. Results

5.1. Results of conventional models for binary classification

As observed in Table 3, that the models perform well with SMOTE with CNN having highest accuracy of 98. %.

5.2. Results of conventional models for multi-class classification

As observed in Table 4, that the models perform well with SMOTE as observed in the case of binary class data with CNN having highest accuracy of 94.4. %.

Table 3
Results of binary classification via conventional DL pipeline.
(a) Before SMOTE (b) After SMOTE.

Deep learning algorithm	Before SMOTE			After SMOTE		
	Precision	Recall	Accuracy	Precision	Recall	Accuracy
CNN	0.940	0.939	0.951	0.976	0.987	0.984
LSTM	0.929	0.926	0.926	0.955	0.967	0.978

Table 4
Results of multi-class classification via conventional DL pipeline.
(a) Before SMOTE (b) After SMOTE.

Deep learning algorithm	Before SMOTE			After SMOTE		
	Precision	Recall	Accuracy	Precision	Recall	Accuracy
CNN	0.920	0.930	0.915	0.96	0.96	0.944
LSTM	0.835	0.896	0.940	0.938	0.941	0.943

5.3. Results of AutoDL models for binary classification

As observed in Table 5, that the model performs well with SMOTE with highest accuracy of 96.6%.

5.4. Results of AutoDL models for multi-class classification

As observed in Table 6, that the model performs well with SMOTE with highest accuracy of 98.4%.

6. Discussion

Korattur lake water data from Chennai, Tamil Nadu, India was used to execute the CNN, LSTM and AutoKeras models for prediction of water quality. How the data is processed by the CNN, LSTM and AutoKeras is explained in the Section 4. The internal working of the conventional models is a black box which is taken care by the deep learning library called Keras. Comparison of conventional and AutoDL techniques were done on binary and multi-class data set and the evaluation of the models were done based on the metrics such as accuracy, precision and recall. Deep Learning algorithms such as Artificial Neural Network (ANN), Recurrent Neural Network (RNN) and Long Short Term Memory (LSTM) were explored for binary and multi class classification of Korattur Lake water data (Prasad et al., 2020). ANN performed better for binary class data, RNN and LSTM performed better for multi class data. ANN yielded 86% (binary) and 77% (multi class), RNN yielded 87% (binary) and 89% (multi class), LSTM yielded 92% (binary) and 94% (multi class). While Convolution Neural Network (CNN) is explored in the current research work yielded 95% (binary) and 91% (multi class). The results were shown in the following graphs.

Table 5
Results of binary classification via auto DL model.

Auto DL model	Before SMOTE			After SMOTE		
	Precision	Recall	Accuracy	Precision	Recall	Accuracy
AutoKeras	0.863	0.894	0.894	0.971	0.966	0.966

Table 6
Results of multi-class classification via auto DL model.

Auto DL model	Before SMOTE			After SMOTE		
	Precision	Recall	Accuracy	Precision	Recall	Accuracy
AutoKeras	0.567	0.725	0.725	0.985	0.984	0.984

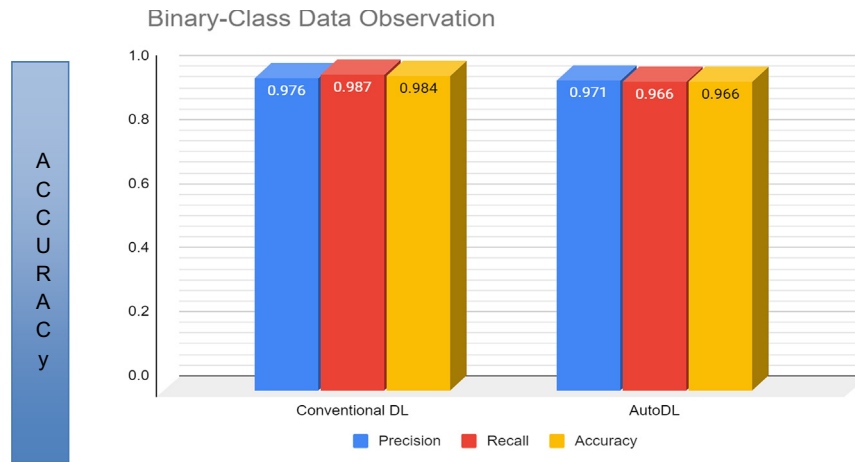


Fig. 7. Bar chart comparison of DL/AutoDL techniques for Binary classification.

6.1. Model evaluation of conventional DL and AutoDL methods for binary classification

As perceived in Fig. 7, conventional DL performs better than AutoDL. One must notice that conventional DL values plotted here consist of the output of the champion model, CNN, which showed improved accuracy over LSTM. The accuracy of conventional DL is 1.8% higher than that of AutoDL with respect to binary class data of water samples. For binary classification, the CNN provided 98.4% accuracy which is higher than the AutoDL model yielding 96.6%. Precision is the ratio between the True Positives and all the positives. The recall is the measure of our model correctly identifying True Positives. Recall also gives a measure of how accurately our model can identify the relevant data. The precision obtained using both conventional and AutoKeras model is 97%. Recall is obtained as 96.6% using AutoKeras and 98.7% using CNN model. The conventional models outperformed the AutoKeras model.

6.2. Model evaluation of conventional DL and AutoDL methods for multi-class classification

As perceived in Fig. 8, that conventional DL performs better than AutoDL. One must notice that conventional DL values plotted here consist

of the output of the champion model, CNN, which showed improved accuracy over LSTM. The accuracy of conventional DL is 1% higher than that of AutoDL for multiclass water data. For multi class classification, the CNN provided 99.4% accuracy which is higher than the AutoDL model yielding 98.4%. The precision obtained using conventional model is 96% and AutoKeras model is 98.5%. Recall is obtained as 96.6% using AutoKeras and 98.7% using CNN model. The conventional models outperformed the AutoKeras model. The accuracy values provide the efficiency of the deep learning model in prediction. Precision, Recall and Accuracy are the metrics for evaluating the prediction models.

7. Conclusion

From the results of the conventional and AutoDL techniques applied over water quality analysis it was inferred that the AutoDL is a new advancement in this field of research. AutoDL model is as competent as conventional DL models, with niches for improvement identified.

7.1. Limitations and scope for future enhancement

It is noticed that data have a significant impact on both DL and AutoDL models in the initial stages. The accuracy was improved after using SMOTE technique. AutoDL consumes very less time compared to conventional DL

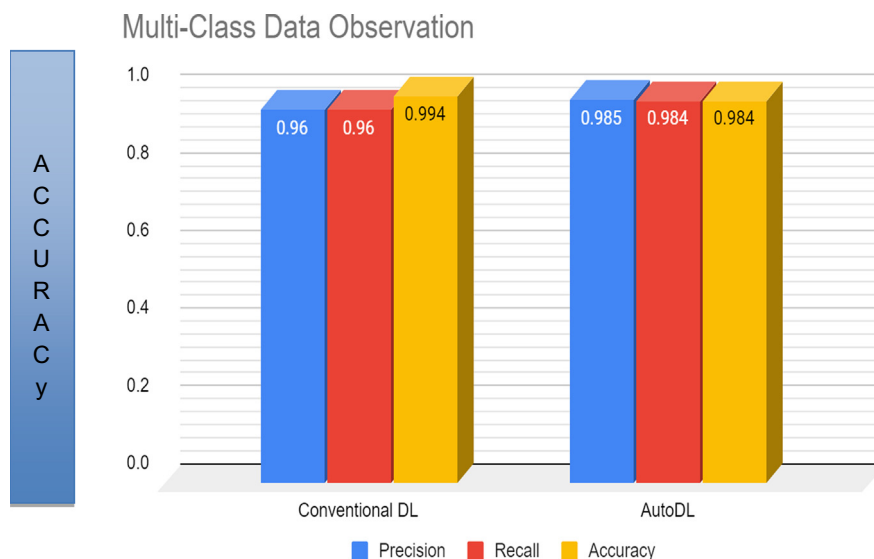


Fig. 8. Bar chart comparison of DL/AutoDL techniques for Multi-class classification.

models. It can also be identified that the performance gap is considerably large between DL and Auto DL. There is much scope for improvement in the AutoDL arena.

AutoDL is appreciably developed for image and textual data and not tabulated structural data. It leaves room for improvement, such as in the case of Water Quality Analysis.

On the performance front, models performed better on Binary Class data than Multi Class data in every experiment, which may be reasoned with the class imbalance that is more prominent in the multi-class data than the binary class data.

As observed from results, AutoKeras has a good infrastructure to support structured/tabulated data but the documentation and the resources for general understanding of the library's functions are poor. Thus, it can be concluded that the right direction for AutoDL to proceed would be to improve malleability across datasets and overcome the same for it to be advantageous for Water Quality Analysis.

CRedit authorship contribution statement

D. Venkata Vara Prasad: Investigation, Methodology, Data curation, Resources, Writing – original draft. **Lokeswari Y. Venkataramana:** Investigation, Methodology, Data curation, Resources, Writing – original draft. **P. Senthil Kumar:** Conceptualization, Methodology, Validation, Supervision. **G. Prasannamedha:** Investigation, Formal analysis, Data curation. **S. Harshana:** Data curation, Resources, Writing – original draft. **S. Jahnavi Srividya:** Data curation, Resources, Writing – original draft. **K. Harrine:** Data curation, Resources, Writing – original draft. **Sravya Indraganti:** Data curation, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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